

Term Test 1 Review

Agenda

- Built in data types: `int`, `float`, `str`, `bool`
- Print vs Return
- String Indexing/methods
- Functions
- File reading/writing
- Control-flow statements
 - `if/elif/else` statements
 - `while` statements

Built-in Data Types

What is the output of the following code?

(a) 16

(b) 33

(c) 101013

(d) 1010103

(e) Error

```
num = str(10) * 3  
print(int(num) + 3)
```

Print vs Return

What is the output of the following code?

(a) 0

(b) None

(c) Good evening

(d) Good evening

None

(a) 0

None

```
def foo():  
    if bool(1):  
        print("Good evening")  
    else:  
        return 0  
  
bar = foo()  
print(bar)
```

String Indexing & While Loops

What is the output of the following code?

(a) 1

(b) APS106

(c) APS360

(d) None

(e) Error

```
def cmp():  
    word1 = "APS106"  
    word2 = "APS360"  
  
    i = 0  
    while i < 6:  
        c1 = word1[i]  
        c2 = word2[i]  
  
        if ord(c1) < ord(c2):  
            return word1  
        elif ord(c1) > ord(c2):  
            return word2  
  
        i += 1  
  
course = cmp()  
print(course)
```

if-statements

```
1 ▾ if eggs % 12 == 0:  
2     return False  
3 ▾ else:  
4     return True
```

Select statements equivalent to the statements above:

- (a) `return eggs%12 == 0`
- (b) `return not eggs % 12 == 0`
- (c) `return not (eggs % 12 != 0)`
- (d) `return eggs % 12 != 0`

If-statements and variables

- What are the values of the variables x , y , and z *after* line 40 this program?

(a) 1.0, 0, 1.05

(b) 1.0, -78, 1.0

(c) 1, -98, 1

(d) 1.0, -98, 1.0

```
30 x = 2.1
31 y = 88
32 z = x // 2
33
34 if x >= 2:
35     y += 10
36 else:
37     y -= 10
38
39 y = -y
40 x = (z ** 3) % 4
```

If-statements and functions

- What line below is equivalent to `mystery_func(x, y, z)`?

(a) `min(x, y, z)`

(b) `max(x, y, z)`

(c) `min(x-z, z-y, 0)`

(d) `min(x-z, z-y)`

```
def mystery_func(x, y, z):  
    if x - z >= 0 and z - y <= 0:  
        return z  
    elif x >= y:  
        return y  
    else:  
        return x
```

Strings and if-statements

- What is printed when the code below is run?

```
course = 'aps106'
new_str = ''

i = 0
while i < len(course):
    if i % 2 == 0:
        new_str = new_str + course[i].upper()
    else:
        new_str = new_str + course[i-1] * 2

    i += 1

print(new_str)
```

Write the code: Inverse tanh

- The inverse hyperbolic tangent function ($\tanh^{-1}(x)$) can be estimated using the following series:

$$\sum_{n=1}^N \frac{x^{2n-1}}{2n-1} = x + \frac{x^3}{3} + \frac{x^5}{5} + \frac{x^7}{7} + \dots$$

where $-1.0 < x < 1.0$ and N is the number of terms used in the approximation.

- **Part 1:** Write a function that takes two parameters: x and N and computes and returns the approximation to the inverse hyperbolic tangent function. Do not use the `math` module.
- **Part 2:** Write a program that prompts a user for values of x and N and then calls the function from part 1. If the user enters invalid values for x (≥ 1 or ≤ -1) or N (non-positive integer), ask the user to enter the values again until they are valid. Finally, print the results.

Write the code: Customer Question Ratio

- A company has collected customer reviews after launching a new product. The reviews are stored in a text file, with one review per line.
- The support team wants to know what percentage of the reviews are questions, since questions usually indicate customers who need assistance.
- Write a Python program that:
 - Opens the file **feedback.txt** and reads it line by line using a while loop. Each line represents one review.
 - Counts how many reviews contain a question. A review is considered a question if the line contains the character "?".
 - After reading the entire file, compute the percentage of reviews that are questions and prints the result rounded to three decimal places.

Write the code: Operation create more data

- Having learned that AI models are running out of data on the internet, you decide to record your conversations to produce more data to sell.
- To create your prototype program that will automatically transcribe your conversations, you decide to start on the part of the code that will write your conversations to a file.
- **Task:**
 - Write a program that repeatedly asks a user to enter some text input and then write the text to a file.
 - The program should end when the user enters “*DONE*”
 - The name of the file should be *name_date_convo_log.txt* where name and date are provided by the user at the beginning of the program

Example

Enter a name: Ben

Enter a date (YYYYMMDD): 20260210

Enter conversation text (or *DONE* to finish):
Wow last week's APS106 lab was so fun

Enter conversation text (or *DONE* to finish):
Yeah! I always do the APS106 labs before my
other assignments

Enter conversation text (or *DONE* to finish):
DONE

File Ben_20260210_convo_log.txt contents

Wow last week's APS106 lab was so fun
Yeah! I always do the APS106 labs before
my other assignments